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Fundamentals of Computer Vision

Three-Class Brain CT Stroke Classification Using VGG16 and ResNet50

Computer Vision Application:
Medical Image Classification

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Dataset: TEKNO21 Brain CT Stroke

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Abstract

Stroke is a time-sensitive neurological condition in which rapid differentiation between ischemic and hemorrhagic stroke is essential. This project investigates automatic three-class classification of brain computed tomography images into normal, ischemic stroke, and bleeding classes using the TEKNO21 Brain CT Stroke dataset.

A preprocessing pipeline was developed to remove unnecessary background, identify and crop the head region, preserve anatomical proportions through square padding, and resize images to 224×224 pixels. Two preprocessing variants were evaluated: one without contrast enhancement and another using Contrast Limited Adaptive Histogram Equalization (CLAHE). Transfer learning was applied using ImageNet-pretrained VGG16 and ResNet50 convolutional neural networks. Both networks were trained in two stages: initial training of a new classification head followed by fine-tuning of the final convolutional or residual stage.

Four experiments were compared. VGG16 achieved 87.98% accuracy and a macro F1-score of 82.16% with both preprocessing variants. Its reported metrics and confusion matrix were unchanged after CLAHE enhancement. ResNet50 with CLAHE achieved 88.88% accuracy and a macro F1-score of 84.11%. The best result was obtained using ResNet50 without CLAHE, which achieved 90.78% accuracy, 86.00% balanced accuracy, and a macro F1-score of 87.06%. The results indicate that ResNet50 provided better recognition of the minority stroke classes. CLAHE produced no measurable improvement for VGG16 and reduced ResNet50 performance under the selected experimental settings.

1. Introduction

1.1. Problem Background

Stroke occurs when blood flow to part of the brain is interrupted or when a blood vessel inside or around the brain ruptures. The two major stroke types are ischemic stroke, which is caused by reduced or blocked blood flow, and hemorrhagic stroke, which is caused by bleeding. These conditions require different medical management; consequently, differentiating between them is an important stage in clinical assessment.

Non-contrast computed tomography (CT) is commonly used in emergency brain imaging because it is fast and can reveal intracranial bleeding. Acute hemorrhage often appears relatively hyperdense, while ischemic changes may appear hypodense or may initially be subtle. These visual differences make brain CT classification a suitable computer vision problem, but the task remains challenging because ischemic findings can be faint, abnormalities can occupy only a small region, and normal anatomical variations can resemble pathological patterns.

The TEKNO-21 stroke dataset was prepared using anonymized medical images and expert radiologist annotations. The dataset includes hemorrhagic, ischemic, and non-stroke images and was developed to support artificial intelligence research in stroke analysis [1].

1.2. Computer Vision Application

The selected application is **supervised medical image classification**. The goal is to learn a function

$$f(\mathbf{x}) \rightarrow y,$$

where $\mathbf{x}$ represents a brain CT slice and y belongs to one of three classes:

$$y \in \{\text{Normal}, \text{Ischemia}, \text{Bleeding}\}.$$

The model performs classification at the image level. It does not segment the lesion, identify its exact location, or provide a patient-level medical diagnosis.

1.3. Project Objectives

The main objectives of the project were:

1. Develop a reproducible preprocessing pipeline for brain CT images.
2. Remove unnecessary black background while retaining the full head.
3. Compare images processed with and without local contrast enhancement.
4. Apply transfer learning using VGG16 and ResNet50.
5. Compare the effect of CLAHE on both architectures.
6. Address the class imbalance present in the dataset.
7. Evaluate the models using class-sensitive metrics rather than accuracy alone.
8. Analyze the major classification errors and identify the strongest experimental configuration.

2. Dataset

2.1. TEKNO21 Brain CT Stroke Dataset

The project used the processed PNG version of the TEKNO21 Brain CT Stroke dataset. The processed release contains 6,653 brain CT images divided into three categories:

Table 1: Class distribution in the processed TEKNO21 dataset.

Class	Number of images	Approximate percentage
Normal	4,428	66.56%
Ischemia	1,131	17.00%
Bleeding	1,094	16.44%
Total	6,653	100.00%

The distribution shows a clear class imbalance. Approximately two-thirds of the images belong to the normal class, while ischemia and bleeding each represent approximately one-sixth of the dataset. Therefore, overall accuracy alone could be misleading because a model may obtain a high score by favoring the majority class.

2.2. Data Splits

Three CSV files were used to define the training, validation, and testing sets. Each file contained:

- the image path,
- the text class name,
- the numeric class label.

The same split files were used for all experiments to ensure a fair comparison. The class mapping was:

$$0 \rightarrow \text{Normal}, \qquad 1 \rightarrow \text{Ischemia}, \qquad 2 \rightarrow \text{Bleeding}.$$

The held-out test set contained 998 images, distributed as follows:

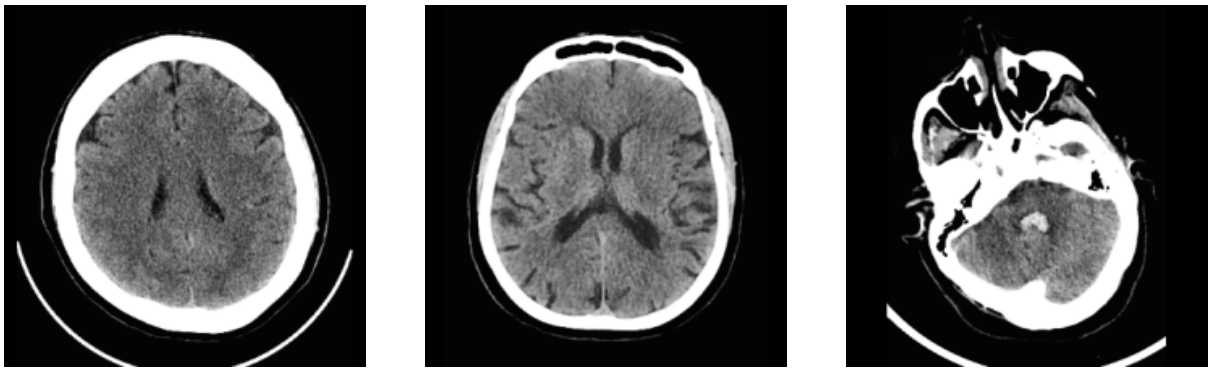
Table 2: Class distribution of the test set.

Class	**Test images**	**Percentage**
Normal	664	66.53%
Ischemia	170	17.03%
Bleeding	164	16.43%
Total	998	100.00%

The implementation checked that the same exact image path did not appear in more than one split. However, patient-level identifiers were not available in the processed CSV files, so the possibility of different slices from the same patient appearing in different splits could not be completely excluded.

2.3. Representative Images

Figure 1 presents representative preprocessed CT slices from the three dataset classes. The examples illustrate variation in anatomical level, ventricular appearance, image intensity, and visible abnormalities.



(a) Normal **(b)** Ischemia **(c)** Bleeding

Figure 1: Representative brain CT images from the three dataset classes after cropping, square padding, resizing to 224×224, and without CLAHE enhancement.

3. Methods

3.1. Overview

The full experimental workflow consisted of the following stages:

1. Load the processed CT image.
2. Detect the foreground head region.
3. Crop unnecessary black background.
4. Pad the cropped image to a square.
5. Resize the image to 224×224.
6. Optionally apply CLAHE.
7. Apply training-only data augmentation.
8. Train an ImageNet-pretrained CNN using transfer learning.
9. Fine-tune the final feature-extraction stage.
10. Evaluate the selected checkpoint on the held-out test set.

3.2. Grayscale Handling

The TEKNO21 PNG images were treated as grayscale brain CT images. During model loading, TensorFlow explicitly decoded every image as a single-channel grayscale image:

$$\mathbf{x} \in \mathbb{R}^{224\times224\times1}.$$

VGG16 and ResNet50 pretrained weights expect three-channel input. Therefore, the single grayscale channel was copied into three identical channels:

$$\mathbf{x}_{RGB} = [\mathbf{x}, \mathbf{x}, \mathbf{x}] \in \mathbb{R}^{224\times224\times3}.$$

This operation does not create new color information. It only makes the image shape compatible with the pretrained networks.

3.3. Head Detection and Background Cropping

A large black region surrounds the head in many CT images. Keeping the entire background would waste input resolution and allow the network to learn irrelevant border patterns. A head-detection mask was therefore generated.

Pixels with values greater than 5 were treated as foreground:

$$M(x, y) = \begin{cases} 255, & I(x, y) > 5, \\ 0, & \text{otherwise.} \end{cases}$$

A 5×5 square structuring element was used for morphological processing:

$$K = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Morphological closing was applied for two iterations to fill small black gaps and connect nearby foreground regions. Morphological opening was then applied for one iteration to remove isolated foreground noise.

These operations were applied only to the binary head-detection mask. They were not applied directly to the medical CT intensities.

Connected-component analysis with 8-connectivity was used to identify separate foreground regions. The largest connected foreground component was assumed to represent the head. Components occupying less than 5% of the full image area were rejected to reduce the risk of selecting text, markers, or small artifacts.

A bounding box was calculated around the selected component. An 8% margin was added in both spatial directions to reduce the risk of removing skull-adjacent abnormalities.

3.4. Square Padding and Resizing

After cropping, the image could be rectangular. Directly resizing a rectangular image into 224×224 would distort the anatomical proportions of the head. Therefore, black padding was added to the shorter dimension to create a square image.

For an image with height h and width w, the square dimension was

$$s = \max(h, w).$$

The cropped image was centered in an $s \times s$ black canvas and then resized to

$$224 \times 224.$$

Area interpolation was used when reducing image size, while linear interpolation was used when enlarging an image.

3.5. Contrast Enhancement Experiment

Two preprocessing variants were evaluated for both VGG16 and ResNet50.

3.5.1 No-contrast version

The no-contrast dataset included cropping, square padding, and resizing but did not apply additional local histogram enhancement.

3.5.2 CLAHE version

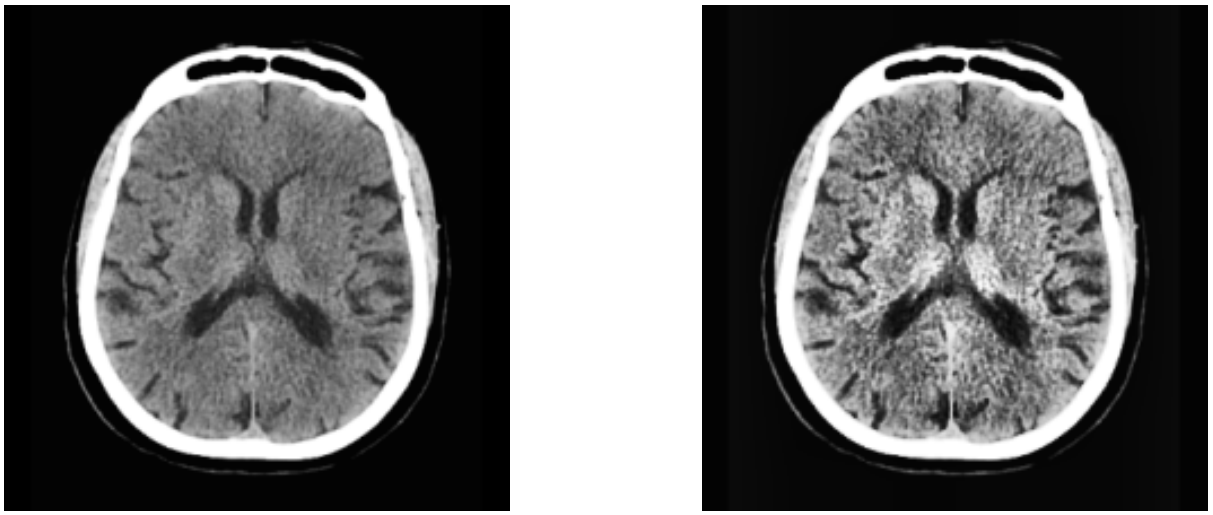
The contrast-enhanced dataset applied Contrast Limited Adaptive Histogram Equalization after resizing. CLAHE divides an image into local tiles, equalizes the intensity distribution within each tile, and limits excessive contrast amplification [4].

The selected parameters were:

Table 3: CLAHE parameters.

Parameter	Value
Clip limit	2.0
Tile grid size	8×8

CLAHE was included as an ablation experiment rather than being assumed to improve performance. Local contrast enhancement can reveal weak patterns, but it can also amplify image noise or alter intensity relationships.



(a) Without CLAHE **(b)** With CLAHE

Figure 2: Visual comparison of the same CT slice before and after CLAHE enhancement using a clip limit of 2.0 and an 8×8 tile grid.

3.6. Data Augmentation

Data augmentation was applied only to the training images. Validation and test images were not augmented.

The augmentation pipeline contained:

- horizontal flipping,
- random rotation with a factor of 0.02, approximately $\pm 7.2°$,
- horizontal and vertical translation of up to 5%,
- random zoom of approximately $\pm 10\%$.

The augmentations were intentionally mild to preserve realistic brain anatomy. Vertical flips, large rotations, and strong geometric deformations were not used.

3.7. Class Imbalance Handling

Balanced class weights were calculated using only the training labels. The weight of class c was defined as

$$w_c = \frac{N}{KN_c},$$

where:

- N is the total number of training samples,
- K is the number of classes,

- N_c is the number of training samples in class c.

This gives greater loss contribution to minority-class examples and reduces the tendency of the model to predict the majority normal class.

3.8. Transfer Learning

Transfer learning was used because the medical dataset was much smaller than ImageNet. Both networks were initialized using ImageNet-trained weights. Their original 1,000-class classification layers were removed by setting `include_top=False`.

The shared classification head consisted of:

1. global average pooling,
2. a fully connected layer with 256 units and ReLU activation,
3. dropout with a probability of 0.50,
4. a three-unit softmax output layer.

For class c, the softmax probability was

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{3} \exp(z_j)}.$$

3.9. VGG16

VGG16 is a deep convolutional architecture based mainly on repeated 3×3 convolutional filters and max-pooling operations [2]. Its simple sequential design provides a strong transfer learning baseline.

During the first training stage, the complete VGG16 feature extractor was frozen. During fine-tuning, only layers whose names began with `block5_` were made trainable.

VGG16 was evaluated using both the no-contrast and CLAHE-enhanced datasets.

3.10. ResNet50

ResNet50 uses residual connections that allow information and gradients to pass through shortcut paths. These connections make deeper networks easier to optimize [3].

During stage one, the complete ResNet50 feature extractor was frozen. During fine-tuning, layers in the final residual stage, identified by names beginning with `conv5_`, were unfrozen. Batch-normalization layers remained frozen to avoid unstable updates to their running statistics on the smaller medical dataset.

ResNet50 was evaluated using both the no-contrast and CLAHE-enhanced datasets.

3.11. Experimental Configurations

The four experiments were:

Table 4: Experimental configurations.

Experiment	Architecture	Preprocessing variant
1	VGG16	No CLAHE
2	VGG16	CLAHE
3	ResNet50	No CLAHE
4	ResNet50	CLAHE

The same training, validation, and test split definitions were used across the four experiments.

3.12. Training Configuration

The same primary training configuration was used across experiments.

Table 5: Training hyperparameters.

Hyperparameter	Value
Input size	$224 \times 224 \times 3$
Batch size	16
Optimizer	Adam [5]
Stage 1 learning rate	1×10^{-4}
Stage 2 learning rate	1×10^{-5}
Maximum stage 1 epochs	20
Maximum stage 2 epochs	10
Loss function	Sparse categorical cross-entropy
Dropout rate	0.50
Random seed	42
Model selection criterion	Minimum validation loss

Sparse categorical cross-entropy was used:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} w_{y_i} \log p(y_i \mid \mathbf{x}_i).$$

3.13. Callbacks and Model Selection

The following callbacks were used:

- **Model checkpointing:** saved the model with the lowest validation loss.

- **Early stopping:** stopped training after six epochs without improvement and restored the best weights.
- **Learning-rate reduction:** multiplied the learning rate by 0.2 after three epochs without validation-loss improvement.
- **CSV logging:** stored epoch-level loss, accuracy, and validation metrics.

The best checkpoint from stage one was compared with the best checkpoint from stage two. The checkpoint with the lower validation loss was selected for final test evaluation.

3.14. Evaluation Metrics

Because the dataset was imbalanced, several evaluation metrics were used.

3.14.1 Accuracy

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}.$$

3.14.2 Precision

$$\text{Precision} = \frac{TP}{TP + FP}.$$

3.14.3 Recall

$$\text{Recall} = \frac{TP}{TP + FN}.$$

3.14.4 F1-score

$$F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

3.14.5 Balanced Accuracy

Balanced accuracy is the average recall across all classes:

$$\text{Balanced Accuracy} = \frac{1}{K}\sum_{c=1}^{K} \text{Recall}_c.$$

Macro averages give equal importance to each class, while weighted averages account for class support. Confusion matrices were also examined to identify specific error patterns between normal, ischemia, and bleeding.

4. Results

4.1. Overall Model Comparison

Table 6 compares the four experiments.

Table 6: Overall test performance of the four experiments.

Experiment	Accuracy	Balanced Accuracy	Macro Precision	Macro F1	Weighted F1
VGG16, no CLAHE	87.98%	78.52%	87.31%	82.16%	87.40%
VGG16, CLAHE	87.98%	78.52%	87.31%	82.16%	87.40%
ResNet50, CLAHE	88.88%	82.92%	85.96%	84.11%	88.88%
ResNet50, no CLAHE	**90.78%**	**86.00%**	**88.23%**	**87.06%**	**90.70%**

ResNet50 without CLAHE achieved the strongest value for every listed overall metric. It correctly classified 906 of the 998 test images.

The two VGG16 experiments produced the same reported metrics. Therefore, CLAHE did not create a measurable change in VGG16's aggregate test performance under the selected settings.

4.2. Per-Class Results

Table 7: Per-class precision, recall, and F1-score.

Experiment	Class	Precision	Recall	F1-score
VGG16, no CLAHE	Normal	88.40%	97.59%	92.77%
	Ischemia	79.55%	61.76%	69.54%
	Bleeding	93.98%	76.22%	84.18%
VGG16, CLAHE	Normal	88.40%	97.59%	92.77%
	Ischemia	79.55%	61.76%	69.54%
	Bleeding	93.98%	76.22%	84.18%
ResNet50, CLAHE	Normal	92.65%	94.88%	93.75%
	Ischemia	71.20%	77.06%	74.01%
	Bleeding	94.03%	76.83%	84.56%
ResNet50, no CLAHE	Normal	93.25%	95.63%	94.42%
	Ischemia	80.72%	78.82%	79.76%
	Bleeding	90.73%	83.54%	86.98%

The strongest improvement produced by ResNet50 without CLAHE occurred in the ischemia class. Its ischemia recall was 78.82%, compared with 61.76% for both VGG16 experiments. Bleeding recall also increased from 76.22% to 83.54%.

Both VGG16 experiments obtained the highest normal recall, 97.59%, but this was accompanied by much weaker ischemia sensitivity. This indicates that VGG16 predicted the majority normal class too frequently.

4.3. VGG16 Without CLAHE

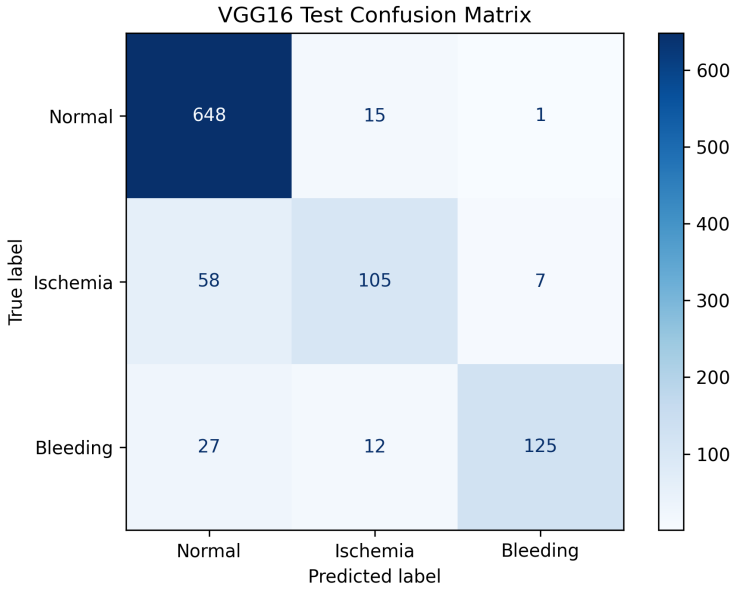


Figure 3: Test confusion matrix for VGG16 using the no-CLAHE dataset.

VGG16 without CLAHE correctly classified 648 normal images, 105 ischemia images, and 125 bleeding images.

Its largest error was ischemia being classified as normal. This occurred in 58 of the 170 ischemia images, corresponding to 34.12% of the ischemia test samples. This error pattern explains why the model's balanced accuracy was considerably lower than its overall accuracy.

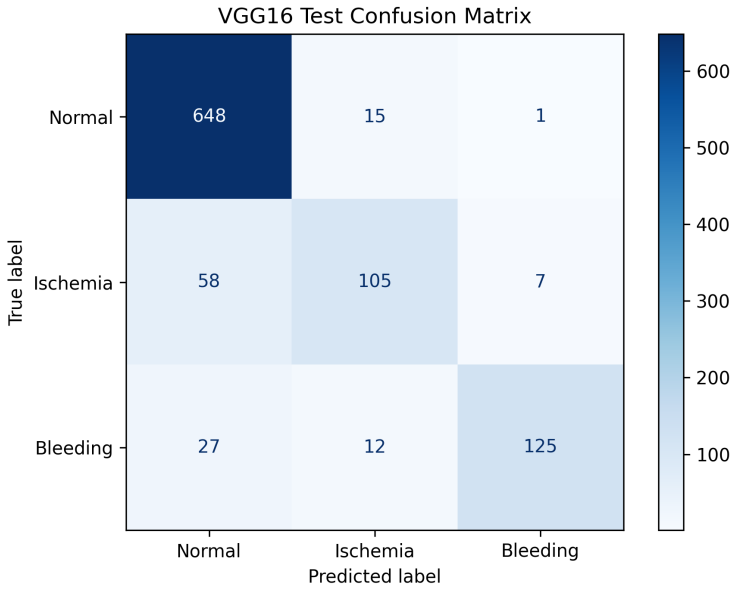


Figure 4: Test confusion matrix for VGG16 using CLAHE-enhanced images.

VGG16 with CLAHE also correctly classified 648 normal images, 105 ischemia images, and 125 bleeding images. Its confusion matrix was identical to the no-CLAHE VGG16 confusion matrix.

The reported class precision, recall, F1-scores, overall accuracy, balanced accuracy, and confusion matrix were unchanged. Therefore, no measurable difference between the two VGG16 preprocessing variants was observed on the held-out test set.

4.4. ResNet50 With CLAHE

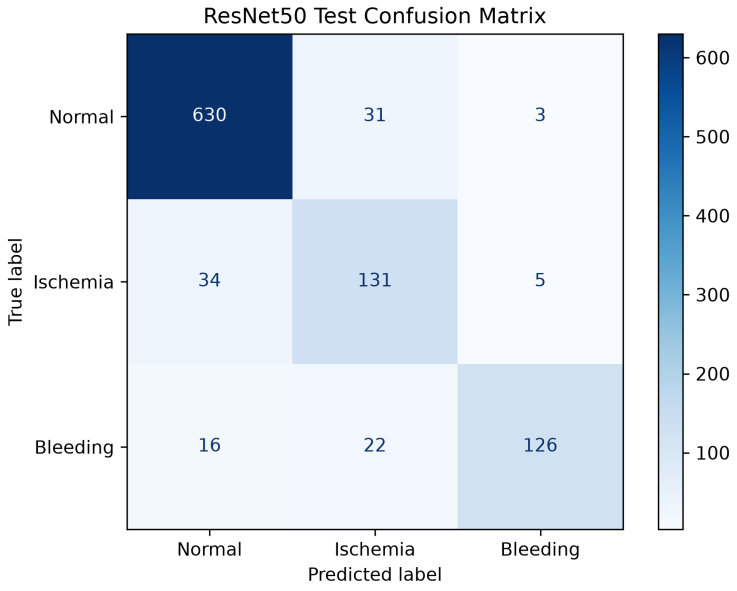


Figure 5: Test confusion matrix for ResNet50 using CLAHE-enhanced images.

ResNet50 with CLAHE correctly classified 630 normal images, 131 ischemia images, and 126 bleeding images, for a total of 887 correct predictions.

Compared with VGG16, it reduced the number of ischemia images predicted as normal from 58 to 34. However, 22 bleeding images were classified as ischemia, indicating increased confusion between the two stroke classes.

4.5. ResNet50 Without CLAHE

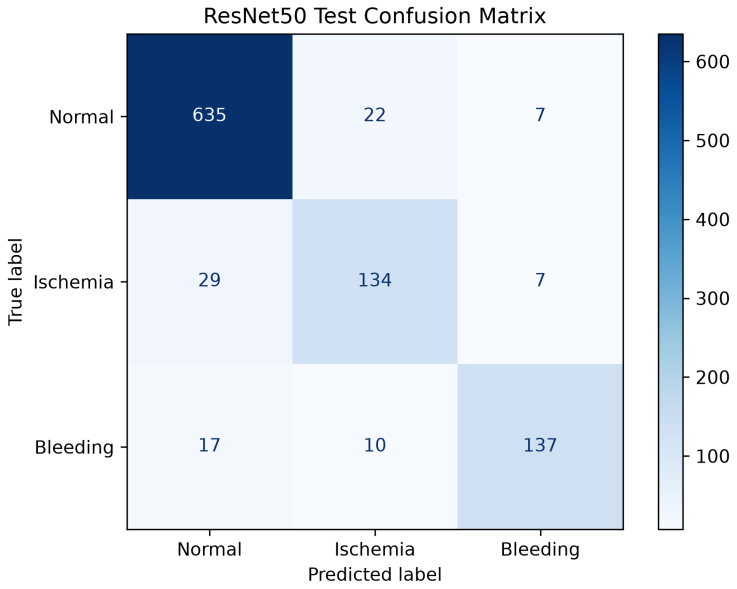


Figure 6: Test confusion matrix for ResNet50 using images without CLAHE.

The best-performing model correctly classified:

- 635 of 664 normal images,
- 134 of 170 ischemia images,
- 137 of 164 bleeding images.

Only 29 ischemia images were predicted as normal, compared with 58 for both VGG16 experiments. The number of bleeding images predicted as ischemia was also reduced to 10, compared with 22 in the ResNet50 CLAHE experiment.

4.6. Effect of CLAHE

The effect of CLAHE differed between the two architectures.

Table 8: Effect of CLAHE on VGG16 and ResNet50 performance.

Architecture	Metric	CLAHE	No CLAHE	No-CLAHE difference
VGG16	Accuracy	87.98%	87.98%	0.00 points
	Balanced accuracy	78.52%	78.52%	0.00 points
	Macro F1	82.16%	82.16%	0.00 points
ResNet50	Accuracy	88.88%	90.78%	+1.90 points
	Balanced accuracy	82.92%	86.00%	+3.08 points
	Macro F1	84.11%	87.06%	+2.95 points

For VGG16, CLAHE produced no change in the reported metrics or confusion matrix. The two preprocessing variants resulted in the same numbers of correct and incorrect predictions for each class.

For ResNet50, the no-CLAHE version achieved higher recall and F1-score for all three classes, as well as stronger overall performance. However, the CLAHE version obtained slightly higher precision for the bleeding class.

The greatest ResNet50 difference was observed for bleeding recall, which increased from 76.83% with CLAHE to 83.54% without CLAHE, a difference of 6.71 percentage points.

CLAHE increased the number of bleeding images classified as ischemia by ResNet50 from 10 to 22. It also increased normal-to-ischemia confusion from 22 to 31 images.

These results show that the effect of preprocessing can depend on the network architecture. CLAHE did not affect the reported VGG16 evaluation, while it reduced ResNet50 performance. This does not demonstrate that CLAHE is generally unsuitable for CT analysis. It only shows that the selected clip limit, tile size, application order, model configurations, and dataset split did not produce an improvement in these experiments.

4.7. Architecture Comparison

Without CLAHE, ResNet50 outperformed VGG16 by:

- 2.80 percentage points in accuracy,
- 7.48 percentage points in balanced accuracy,
- 4.90 percentage points in macro F1-score,
- 17.06 percentage points in ischemia recall,
- 7.32 percentage points in bleeding recall.

With CLAHE, ResNet50 also outperformed VGG16 by:

- 0.90 percentage points in accuracy,
- 4.40 percentage points in balanced accuracy,
- 1.95 percentage points in macro F1-score.

VGG16 retained a higher normal recall, but its minority-class performance was substantially weaker. The results suggest that the residual architecture learned more transferable and discriminative representations for subtle stroke patterns.

Residual connections may have allowed the deeper model to preserve and refine useful features while reducing the optimization difficulty associated with deeper networks.

4.8. Error Analysis

The four experiments made the following total numbers of errors:

Table 9: Correct and incorrect test predictions.

Experiment	Correct	Incorrect
VGG16, no CLAHE	878	120
VGG16, CLAHE	878	120
ResNet50, CLAHE	887	111
ResNet50, no CLAHE	906	92

The most concerning error in all experiments was stroke being predicted as normal. In the strongest model:

- 29 ischemia images were classified as normal,
- 17 bleeding images were classified as normal.

These errors are more important than normal images being classified as stroke because a false-normal prediction could cause a stroke image to be overlooked in a screening scenario.

The ischemia class remained the most difficult class. Its abnormalities may be subtler than acute bleeding, and some ischemic slices may contain weak or localized changes. The remaining errors show that high overall accuracy does not mean that the model is equally reliable for all classes.

5. Discussion

The experiments demonstrate the importance of evaluating both model architecture and preprocessing choices. ResNet50 outperformed VGG16 under both preprocessing conditions, even though both architectures used the same input size, augmentation methods, classification head, loss function, class weighting approach, and two-stage training strategy.

The two VGG16 runs produced the same reported test metrics and confusion matrix. One possible interpretation is that the features learned by VGG16 were not sufficiently affected by the selected CLAHE transformation to alter the final class decisions. Another possibility is that the selected CLAHE parameters enhanced details that did not influence VGG16's learned decision boundaries. Since only aggregate results were analyzed, this experiment does not establish that the internal probability outputs were necessarily identical.

VGG16 performed strongly on the majority normal class but showed a clear minority-class weakness. Its 87.98% accuracy initially appears competitive, but its balanced accuracy was only 78.52%. The 9.46 percentage-point gap between these metrics demonstrates that the class imbalance affected the apparent performance.

ResNet50 without CLAHE produced a smaller gap between overall accuracy and balanced accuracy. Its accuracy was 90.78%, while balanced accuracy was 86.00%. The model therefore produced more even class-level performance.

The comparison between CLAHE and no CLAHE was particularly useful. Image enhancement is sometimes applied automatically under the assumption that higher visible contrast will improve classification. The experiments showed that this assumption was not supported by the current results. CLAHE produced no reported improvement for VGG16 and reduced ResNet50 performance.

The original intensity structure may have contained useful information that was partially altered by local equalization. CLAHE may also have enhanced scanner noise, skull edges, or normal tissue texture. The deeper ResNet50 architecture may have been more sensitive to these altered local textures than VGG16.

Nevertheless, the comparison was limited to one CLAHE configuration. The selected values of clip limit 2.0 and tile grid size 8×8 were reasonable starting settings but were not optimized through a systematic parameter search.

6. Limitations

Several limitations should be considered when interpreting the results.

6.1. Potential Patient-Level Leakage

The implementation verified that identical image paths did not appear in more than one split. However, it did not confirm that all slices from one patient were restricted to a single split. If multiple slices from the same patient were distributed across training and testing, the reported test performance could be optimistic.

6.2. Use of Processed PNG Images

The project used 8-bit PNG images rather than original DICOM data. Therefore, original Hounsfield-unit information, rescale parameters, window settings, pixel spacing, and other acquisition metadata were not directly available. A DICOM-based system could use multiple clinically relevant CT windows and retain more intensity information.

6.3. Single Train–Validation–Test Split

The experiments used one predefined split and one principal random seed. Cross-validation or repeated experiments were not conducted. Therefore, variation caused by the exact split or random initialization was not measured.

6.4. Single CLAHE Configuration

Only one CLAHE configuration was evaluated: a clip limit of 2.0 and a tile grid size of 8×8. Other parameter combinations may produce different results. The experiment therefore evaluates the selected CLAHE configuration, not every possible form of local contrast enhancement.

6.5. Image-Level Rather Than Patient-Level Classification

Each CT slice was classified independently. A complete CT study contains a sequence of neighboring slices. Using only one slice removes three-dimensional context and may make subtle ischemic findings more difficult to identify.

6.6. No External Clinical Validation

The models were evaluated only on the selected dataset split. They were not tested on images from a different hospital, scanner, reconstruction method, or patient population. The models must not be interpreted as clinically validated diagnostic systems.

7. Future Work

Future development could include:

1. Creating patient-level training, validation, and test splits.
2. Training directly from DICOM images using Hounsfield-unit conversion and multiple CT windows.
3. Testing additional CLAHE clip limits and tile-grid sizes.
4. Comparing CLAHE application before and after resizing.
5. Evaluating focal loss or class-balanced focal loss.
6. Performing repeated runs or cross-validation.
7. Adding Grad-CAM visualizations to inspect the regions influencing predictions.
8. Using adjacent slices through a 2.5D or 3D architecture.
9. Evaluating DenseNet, EfficientNet, or vision transformer models.
10. Calibrating class probabilities to improve confidence reliability.
11. Testing the final model on an external dataset.

8. Conclusion

This project developed a complete computer vision pipeline for three-class brain CT stroke classification using the TEKNO21 dataset. The pipeline included head-region detection, morphological mask cleanup, background cropping, square padding, resizing, grayscale handling, augmentation, class weighting, transfer learning, and two-stage fine-tuning.

Four experiments were evaluated by combining two architectures with two preprocessing variants. VGG16 achieved 87.98% accuracy, 78.52% balanced accuracy, and a macro F1-score of 82.16% both with and without CLAHE. Its reported metrics and confusion matrix were unchanged by the selected contrast-enhancement procedure.

ResNet50 with CLAHE achieved 88.88% accuracy and a macro F1-score of 84.11%. The strongest model was ResNet50 without CLAHE, which achieved 90.78% accuracy, 86.00% balanced accuracy, and a macro F1-score of 87.06%.

The results show that ResNet50 provided better minority-class recognition than VGG16 under both preprocessing conditions. They also show that contrast enhancement should be experimentally validated rather than assumed to be beneficial. Under the selected settings, CLAHE produced no measurable improvement for VGG16 and reduced ResNet50 performance, particularly bleeding recall.

Although the best model produced promising classification results, ischemia remained the most difficult class, and patient-level leakage could not be fully excluded. The system should therefore be viewed as an academic classification experiment rather than a clinically validated diagnostic tool.

References

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